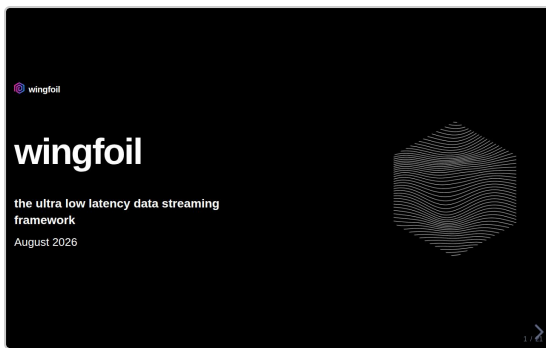


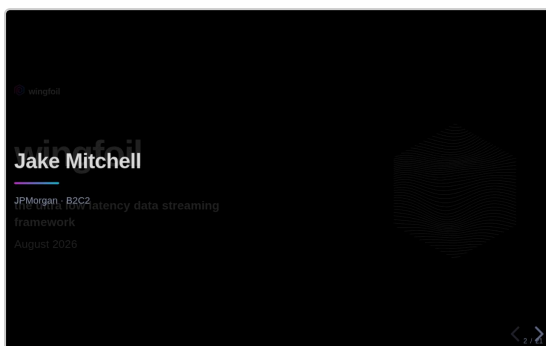
wingfoil — speaker notes

21 slides · printed fallback for the reveal.js speaker view (press S)



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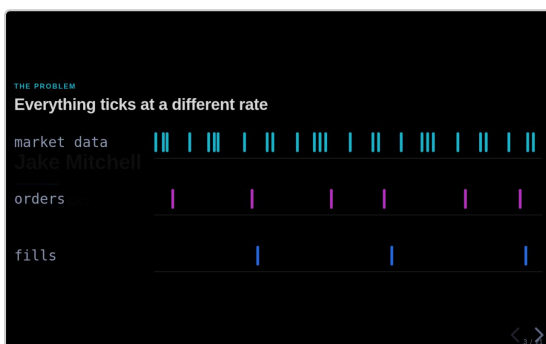
- Thanks for coming. Jake Mitchell. Open source project — **wingfoil**.
- Three parts:
- **1** — background: myself, the problem, and why you need wingfoil to solve it.
- **2** — **Nitro**. The feature we released last week. Why we wanted it. Why it meant rewriting the entire framework. The pattern we found, and what it delivered.
- **3** — what's next.



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Jake Mitchell

- 25 years in financial services.
- Mostly as a quant at JPMorgan — building electronic trading systems for options: across interest rates, FX, precious metals.
- Now at B2C2, working for their crypto options desk.
- I learnt what the most effective patterns were for building these kinds of systems.
- I wanted to bring them into the open source world. And I wanted to use **Rust** to build it.



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THE PROBLEM

Everything ticks at a different rate

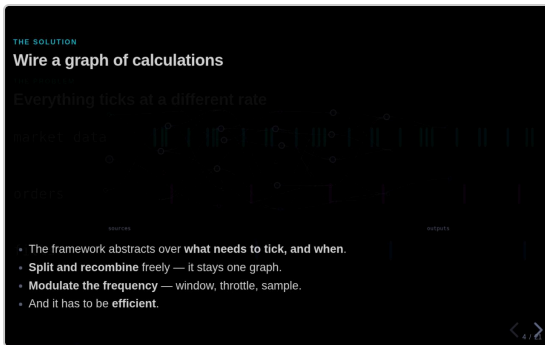
- The problem.
- You're building something that reacts to information that is arriving at **different rates**.
- For example — market data ticking very frequently; orders and fills at a different frequency.
- This could be across tens of thousands of assets.
- And you need to react fast — at micro or even **nanosecond** time scales.

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THE SOLUTION

Wire a graph of calculations

- Wire a graph of calculations.
- Have a framework that **abstracts over what needs to tick at what time**.
- That allows you to split and recombine.
- And to modulate their frequency — for example window or throttle.
- And is **efficient**.



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THE SURFACE

Wire it, build it, run it

- This is what it looks like in practice.
- This example is a simple linear pipeline, with every node ticking on each engine cycle.

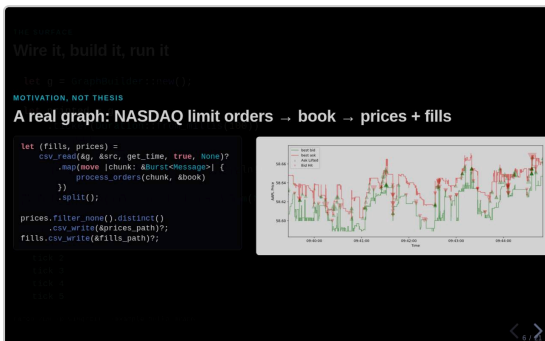


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MOTIVATION, NOT THESIS

A real graph: NASDAQ limit orders → book → prices + fills

- Here is a more interesting example: a **matching engine**.
- Replay limit orders from file.
- Maintain an order book over time.
- Produce top-of-book prices and fills.
- **Three different frequencies of data** — many messages share a timestamp, top of book changes less often, trades are sparser still. Nothing is coalesced or dropped to make them agree.

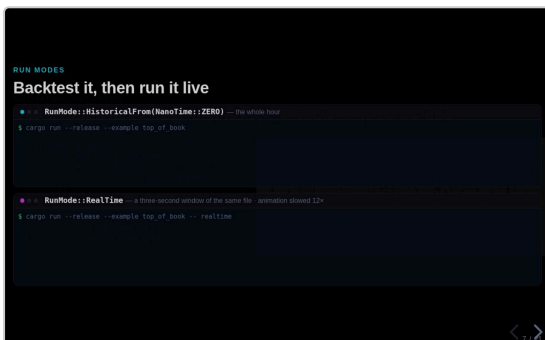


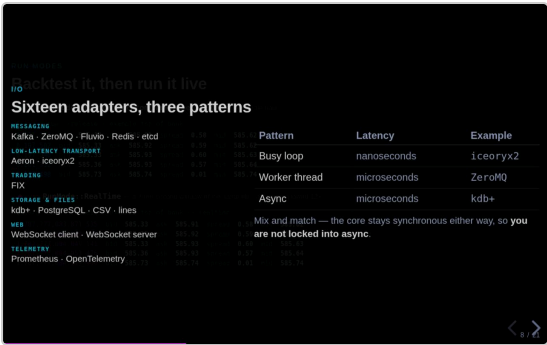
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RUN MODES

Backtest it, then run it live

- Wingfoil supports **realtime** and **historical** modes.
- **Historical** [top] — development · unit tests · integration tests · back test.
- **Realtime** [bottom] — production.
- Same wiring. Same graph. **One line changes** — the run mode.



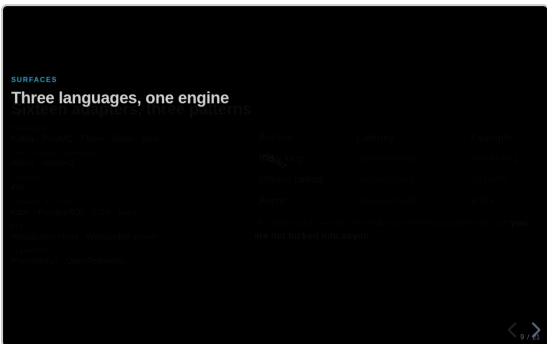


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I/O

Sixteen adapters, three patterns

- We've built a bunch of I/O adapters — FIX, kdb+, iceoryx2, and more.
- They use these three I/O patterns:
 - **Busy spin** — for latency-critical, realtime use cases.
 - **Dedicated worker thread** with waker pattern.
 - **Async** — but only where you need it, at the graph edge. The graph itself is synchronous.
- You can **mix and match** — choose the pattern that works for your use case.

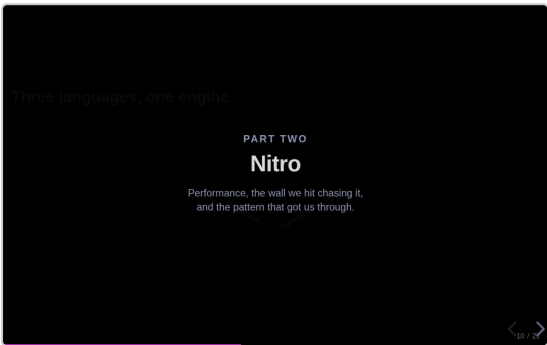


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SURFACES

Three languages, one engine

- **Rust** — the best language to work with.
- Also **Python** bindings, for those that have yet to make the jump.
- **TypeScript** integrations, to stream into the browser.



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PART TWO

Nitro

- That's part one.
- Part two — Nitro.

What we wanted: compile the graph away

- The interpreted engine is fine, but there's overhead I don't love. Per node, per cycle: a **dynamic dispatch** and a **trip through a value slot**.
- On a hot chain that's bigger than the arithmetic you came to do.
- Obvious answer: **generate code**. Straight-line function, state in locals, optimiser across node boundaries. Every ingredient exists.
- So we built it. Wiring at build time, walk the graph, emit a runner. **It failed. Three ways.**
- **1** — had to re-implement every node, because generated code couldn't *call* a node's cycle. Two maps, two filters, nothing keeping them in step.
- **2** — types and closures already gone. You cannot get a closure back out of a `Box<dyn Fn>`. The workaround was the user retyping every closure in a side table. *[Usually gets a laugh. Let it.]*
- **3** — nothing declared how it wanted scheduling, so the engine drove nodes off **allowlists of names**.
- We deleted it. *[Pause.]*
- **The problem was never the generator.** No cleverness in an emitter fixes a design with nothing to emit a call *to*. That's why it meant rewriting the framework.

THE ENGINEERING PROBLEM

What we wanted: compile the graph away

Interpreted, every node pays a **dynamic dispatch** and a **trip through a value slot** — per node, per cycle. On a hot chain that dominates the arithmetic you came to do.

So: **generate code**. One straight-line function, state in locals, the optimiser working across node boundaries.

We built exactly that — ran the wiring from `build.rs`, walked the graph, emitted a runner. It failed three ways.

- 1 It had to re-implement every node. Semantics were welded to `RefCell` fields, so generated code could not call a node's cycle.
- 2 Types and closures were already gone. You cannot get `|x| x * 10` back out of a `Box<dyn Fn>`.
- 3 Scheduling was inferred from names — the engine drove nodes off allowlists.

The problem was never the generator. Node semantics were not *separately callable*.

You write the wiring once, in plain Rust

- You write the wiring once. Plain Rust.
- Real function. Valid, reviewable, readable in a PR.
- Out the other side: interpreted, compiled, nested. **Same tokens.**
- `count` used three times → a shared node. This is a real DAG, not a chain.
- *[Constraint, if asked: wiring must be straight-line. Values and closure logic can be as procedural as you like — the TOPOLOGY can't depend on a runtime value, because compiled monomorphizes one local per node.]*

THE FRONT DOOR

You write the wiring once, in plain Rust

```
wiringfn::ctrl {
  fn odds_evens(g: &graphBuilder) -> Stream<String> {
    let count = g.ticker(PERIOD).count();
    let is_even = count.map(|i| i.is_multiple_of(2));
    let is_odd = is_even.map(|b| !b);
    let odd_str = count.filter(&is_odd).map(|i| format!("{}", i is odd));
    let even_str = count.filter(&is_even).map(|i| format!("{}", i is even));
    let labelled = odd_str.merge(even_str);
    labelled
  }
}
```

Out comes `odds_evens::interpreted()`, `odds_evens::compiled()`, and `odds_evens::nested()` — from the same tokens.

The wiring function stays **valid, reviewable Rust** either way. That property was deliberately never traded away. `count` is read three times, so it is a shared apex node — the macro derives a real DAG, not a chain.

One function. One local per node. No dyn.

- What it turns into.
- Kernel on the stack. One local per node — cfg, state, value. Then a loop.
- *[Point at the call.]* **That's the same cycle function the interpreter calls.** Not a copy. A call.
- Wall number one, dead.
- Error context = the binding name *you* wrote, baked in as a static string. Costs nothing until something breaks.
- Whole expansion is committed. 1,730 lines. Don't take my word for it.

```
...AND THIS IS WHAT IT BECOMES
One function. One local per node. No dyn.

let mut _k = Kernel::new(run_node, run_for);

let mut _cfg_wf_anon_1 = PERIOD;
let mut _state_wf_anon_1 = wf_op.ticker_seed_state(&_cfg_wf_anon_1);
let mut _v_wf_anon_1 = wf_op.ticker_seed_value(&_cfg_wf_anon_1);
// ... one cfg / state / value local per node, 11 nodes ...

let mut _dirty = [false; 11usize];
while _k.begin_cycle(&mut _dirty) {
  let _t_wf_anon_1 = / activation / && {
    let mut _ctx = Ctx::new(&mut _k, &mut _state_wf_anon_1, &mut _v_wf_anon_1, &mut _dirty);
    match wf_op.ticker_cycle(&mut _cfg_wf_anon_1, &mut _state_wf_anon_1, &mut _v_wf_anon_1, &mut _dirty) {
      .context("node 0 (wf_anon_1: ticker) cycle")? {
        Tick::Value(_val) => { _v_wf_anon_1 = _val; true }
        Tick::Silent(_val) => { _v_wf_anon_1 = _val; false }
        Tick::Quiet => false,
      }
    }
  }
  // ... 30 more, emitted in topological order ...
}

originally - 11 lines marked - All 1,730 lines committed verbatim beside the example.
github.com/oxingfall-is/wingfall - examples/core/dual_node/expanded/
```

Semantics became an associated function

- Four associated types and a function.
- Cfg (closures live here) · State (engine owns) · In (typed, per cycle) · Out. Plus a const for scheduling.
- **Notice what's missing.** Nothing says where state lives. Nothing says who calls it.
- It's a free function over explicit arguments.
- *[Return type, if asked: Tick has THREE states. Value ticks downstream, Quiet does nothing, Silent updates the value without ticking — what delay needs.]*

```
THE ONE DECISION EVERYTHING FOLLOWS FROM
Semantics became an associated function

trait Op {
  type Cfg;           // construction-time config; closures live here
  type State;         // engine-owned mutable state
  type In<'a>;        // typed inputs, passed in per cycle
  type Out;           // the produced value
  const ACTIVATION: Activation;

  fn cycle(cfg: &mut Self::Cfg, state: &mut Self::State,
    input: Self::In<'a>, ctx: &mut Ctx<'a>) -> Result<Tick<Self::Out>>;
}

Read what is absent. Nothing here says how state is stored, where inputs
come from, or who calls it.
```

The node changed completely. The wiring barely moved.

- Same node, both ways.
- **Left:** &mut self, peeks a stored upstream handle, writes its own field. Owns its upstreams, its state, its output slot.
- **Right:** same node as an op. Owns **nothing** — everything arrives as an argument.
- Those two are unrecognisable.
- **Now the bottom row** — the wiring the user writes. Before and after.
- **One line different.** You hold a builder and make sources on it.
- We turned the engine inside out, and it cost anyone using it one line.

```
THE SAME NODE, BOTH WAYS
The node changed completely. The wiring barely moved.

legacy — the node owns everything
#(node(active = (upstream), output = value: f64))
and #makeSourceForSourceStream {
  fn cycle(&mut self, _state: &mut graphState)
    -> Result<Out> {
    {
      self.value =
        self.upstream.peek_value() * self.factor;
      Ok(true)
    }
  }
}
owns its upstream handles, its state, its output slot
let count = ticker(Duration::from_millis(10))
  .chain();
count.run(&node; &realTime, &refer::Cycles(100));

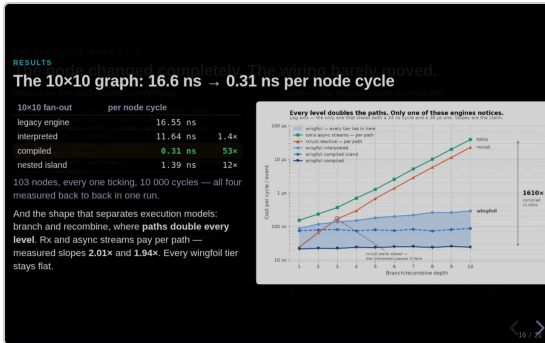
Read what is absent. Nothing here says
come from, or who calls it.

now — the engine owns everything
#(op(build = state, fcount))
and #OpForScale {
  type Cfg = f64;           // config
  type State = ();          // engine-owned
  type In<'a> = (&'a f64);  // passed in
  type Out = f64;
  const ACTIVATION: Activation = Activation::NONE;

  fn cycle(cfg: &mut f64, _state: &mut (),
    input: (&'a f64, &ctx: &mut Ctx<'a>))
    -> Result<TickOfOut> {
    {
      Ok(Tick::Value(input.0 * *cfg))
    }
  }
}
owns nothing — cfg, state and inputs all arrive as arguments
let g = graphBuilder::new();
let count = g.ticker(Duration::from_millis(10))
  .chain();
g.build().run(&node; &realTime, &refer::Cycles(100));
```

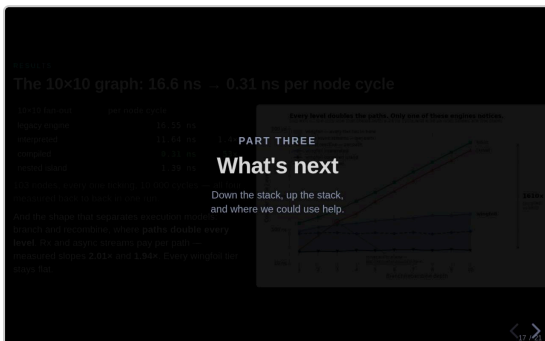
The 10×10 graph: 16.6 ns → 0.31 ns per node cycle

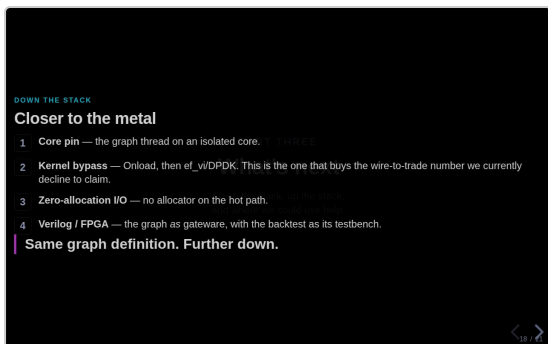
- The 10×10 fan-out — 103 nodes, every one ticking, ten thousand cycles.
- Old engine **16.6 nanoseconds** per node cycle. Compiled: **0.31**. That's **53×**.
- Against the new interpreted engine it's 37×. The island lands between, as it should.
- The one I care about most: **new interpreted beats the OLD engine** — on all eight workloads. Most graphs keep running interpreted, and a rewrite that trades everyday performance for a fast path nobody uses is a bad trade.
- Chart: branch and recombine, paths doubling every level. Rx and async streams pay **per path** — slopes 2.01 and 1.94. We pay per node, so we're flat.
- *[Caveats if pushed: rxrust figure is HALVED from their raw number — their bench emits twice per sample. Tokio target is recursive Box::pin over a ready leaf: honest for per-path propagation, not the best async could do. **The claim is the slope, not the multiplier.**]*
- *[Read the ratios, not the times. Shared cloud VM.]*



What's next

- That's Nitro.
- Last bit — where this goes.



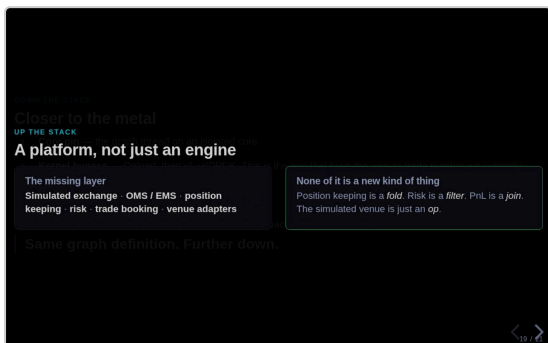


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DOWN THE STACK

Closer to the metal

- **Core pin** — graph thread on an isolated core. Working implementation sits in an example; needs promoting into the runtime.
- **Kernel bypass** — Onload, then raw ef_vi/DPDK. Gets us a wire-to-trade number, which **I won't claim yet because I haven't measured it.**
- **Zero allocation** on the I/O path.
- **Verilog** — the graph as gateware, backtest as testbench. Exploratory. Don't oversell it.
- The point: **none of these change your wiring.** Same graph definition, further down. That's the payoff from part two.

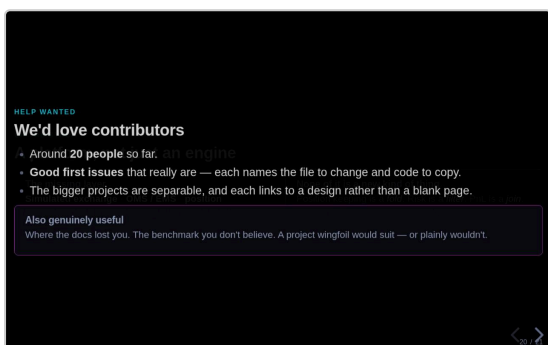


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UP THE STACK

A platform, not just an engine

- Where most of the value is, for most people.
- **Be straight:** wingfoil is a compute engine. It is *not* a trading platform.
- Missing: simulated exchange, OMS, position keeping, risk, booking, more venue adapters.
- Simulator is the hard one — a naive fill-at-touch simulator lies to you, and you'll believe it.
- **The bet:** none of it is a new kind of thing. Position keeping is a *fold*. Risk is a *filter*. PnL is a *join*. The sim venue is just an *op* — RunMode picks.
- Ordinary graph vocabulary. Not a second framework bolted on top.



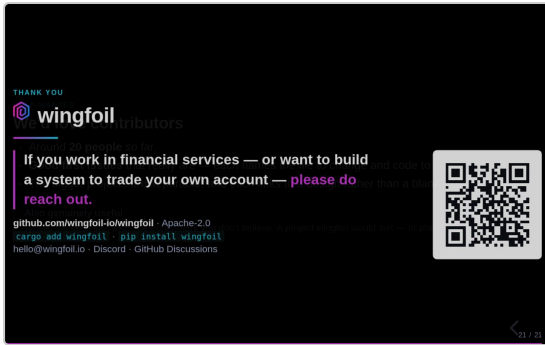
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HELP WANTED

We'd love contributors

- The ask.
- ~20 contributors so far. I'd love more.
- **The good first issues are actually good first issues** — each names the file to change and code to copy. I know that label's been ruined elsewhere.
- Bigger projects are separable — one person each, and each links to a design not a blank page.
- Most useful isn't always code: where the docs lost you · which benchmark you don't believe · a project this would suit, or obviously wouldn't.

wingfoil



- **If you work in financial services — or you're building something to trade your own account — please get in touch.** That's genuinely why I'm up here.
- Repo's on the QR. Thank you.
- *[Leave it up. Don't black the screen — let people scan.]*
- *[If it's quiet: the question I usually get is what happens when the graph shape isn't known until runtime. That's Project Lightning, and it's open.]*